Work Study: Enhancing Productivity Through Methods and Measurement

Sixth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Work Study

Work study may be defined as the analysis of a job for the purpose of finding the preferred method of doing it and also determining the standard time to perform it by the preferred (or given) method. Work study, therefore, comprises of two areas of study: method study (motion study) and time study (work measurement).

What it achieves?

- Enhances efficiency and productivity in processes.
- Achieves cost reduction and time savings.
- Improves workplace ergonomics and safety.

Components of Work Study

- 1. Method Study:

Focus: Optimize methods and reduce unnecessary movements.

2. Work Measurement:

Focus: Determine standard times for tasks.

Method Study Overview

- **Purpose:** To simplify processes and improve workflow.
- **Key Steps:**
- 1. Select the process.
- 2. Record all relevant details.
- 3. Analyze the recorded data.
- 4. Develop a more efficient method.
- 5. Implement and maintain the new method.

Recording Techniques

- **Tools for Recording:**
- - Flow process charts (man, material, machine).
- - String diagrams to visualize movement.
- - Layout designs for workspace optimization.
- **Example:**
- A factory assembly line—analyzing material flow using a flow process chart.

Flow Chart

Distance in mtr.	Symbol	Description	Manpower
	1	Opening of valve manually	
1	⇨	Transportation to PAKONA m/c	1
	2	Filling the spout	1 operator + 1 inspector
1	⇨	Transport by conveyor	
	3	Manually placing in FLOWRAP machine	3 + 1 operator
	□	Inspection	1
1	⇨	Transportation	
	4	Putting in sleeve and counting	2 operator + 1 inspector
1	⇨	Transportation	
	5	Hand sealing	1
1	⇨	Transportation	
	6	Packing	2, 1
	▽	Storage	0
	□	Inspection	0

Distance in mtr.	Symbol	Description	Manpower
	1	Pouch filling on machine	1 operator
1		Transportation	
		Line Inspection	1
	2	Collating	1
	3	Sleeve putting	1
1		Transportation	
	4	Master packing in Cartons	1
1		Transportation	
		Storage	
		Inspection	

Distance in mtr.	Symbol	Description	Manpower
	1	Bottle feeding in pucks	3
1 m		Transport by Conveyor	
	2	Filling	1 operator
	3	Capping	1 operator
		Inspection	2
	4	Puck unloading and inspection	1
	5	Labeling	1 operator
	6	Printing	2 inspectors
1 m		Transportation	
	7	Small Packing (bottle feeding in carton)	2
1 m		Transportation	
	8	Packing (?)	2 operator + 2 inspector
		Transportation	
	9	Putting sleeve and shrinking	5
	10	Master carton packing	1
1 m			
	12	Auto tapping machine	0
		Storage	0

Distance in mtr.	Symbol	Description	Manpower
	1	Automatic filling by machine	1
	⇨	Transportation	
	2	Small carton packing	1+ 1 operator
	□	Inspection	4
	3	Sleeve putting and packing	3
	⇨	Transportation	
	4	Master carton packing	1

Analysis of Work Methods

- **Techniques to Optimize Methods:**
- - Eliminate unnecessary steps.
- - Combine tasks where possible.
- - Rearrange steps for efficiency.
- - Simplify procedures.
- There is a lot of savings by eliminating redundant steps in a process.

Tools Used in Method Study

- - Standardized symbols for recording tasks.
- - Time and motion studies for detailed analysis.
- **Example:** Using a stopwatch to measure task durations.

Frank B. Gilbreth (1868 –1924) and Lillian Gilbreth (1878 –1972)

- Frank Gilbreth and Lillian Gilbreth formed the famous innovative husband-wife team that revolutionized modern Industrial Engineering.
- They have immense contribution in shaping up Industrial Engineering to its modern stage.
- The couple followed in the footsteps of F.W. Taylor and gave concrete shape to Taylor's concept of 'time study' in performing a work.
- The Gilbreths primarily followed the steps of identification, analysis and measurement of basic sequences (or motions) involved in completing a work.
- Frank and Lillian Gilbreth firmly believed that there could be only one best and the most efficient way of doing a work. They conducted extensive studies on the method of doing a work regarding sequence of operations, time needed and idle motion & time.

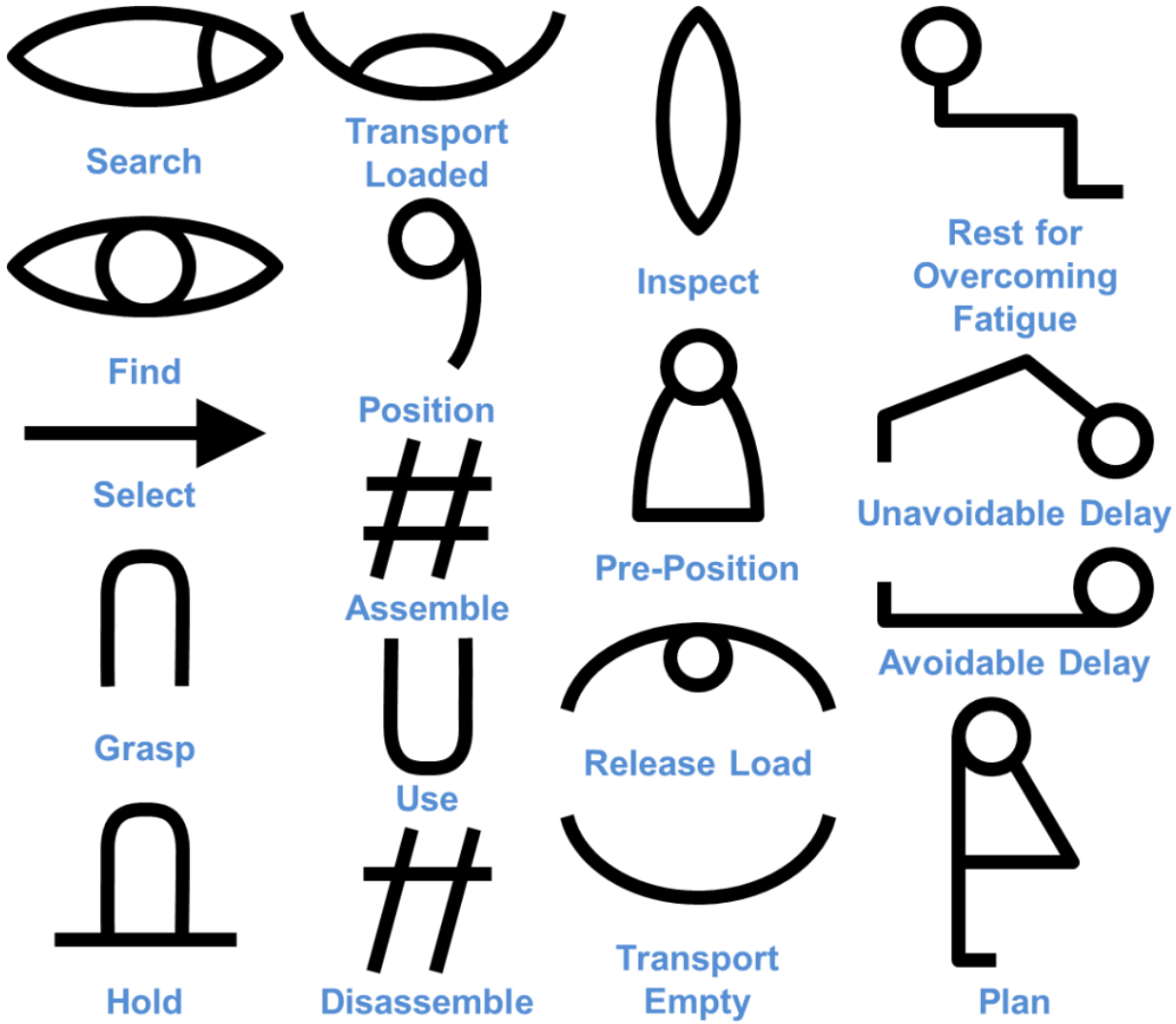
Frank B. Gilbreth and Lillian Gilbreth

- **They were pioneers in using motion pictures for motion study and analysis of work and the workers (http://gilbrethnetwork.tripod.com/bio.html).**
- **Frank Gilbreth was the first to propose that a nurse should help a surgeon by passing surgical instruments to reduce time and unwanted motion during surgery.**
- **He introduced various improved techniques for reducing unwanted motion in the activities of the army and bricklaying methods.**
- **Thus, the Gilbreths introduced 'time and motion study' to identify the most productive and the most efficient method of executing a work.**
- **The Gilbreths tried to implement their rule 'one best way of doing a work for higher efficiency' in their household activities.**
- **Their other innovative contributions towards scientific management, mass production, human factors and ergonomics, and psychology had revolutionary effect on Industrial Engineering, productivity and management practice in an industry.**

Lillian Gilbreth

- Lillian Gilbreth was the first industrial psychologist to integrate psychology with industrial management and obtained a PhD degree on Psychology of Management in 1914.
- The concept of industrial management by considering the personal need of the workers, inter-personal relationships and differences was initiated by Lillian Gilbreth.
- Her management techniques were very successful for improving productivity in an industry.
- She planted the seed of the human relation movement that emerged subsequently in the 1930s.
- Frank Gilbreth and Lillian Gilbreth are immortalized in the classic books ‘Cheaper by the Dozen’ and ‘Belles on Their Toes’ authored by two of the Gilbreths’ children as a tribute to their parents.

THERBLIGS


Search
Transport Loaded
Inspect
Rest for Overcoming Fatigue
Find
Position
Select
Assemble
Pre-Position
Unavoidable Delay
Avoidable Delay
Grasp
Use
Release Load
Hold
Disassemble
Transport Empty
Plan

Work Measurement Overview

- **Purpose:** Establish standard times for tasks.
- **Importance:**
- - Capacity planning.
- - Cost estimation.
- - Fair workload distribution.

Techniques of Work Measurement

- - Time Study
- - Predetermined Motion Time Systems (PMTS)
- - Work Sampling

Time Study Procedure

- 1. Select tasks to measure.
- 2. Break tasks into elements.
- 3. Measure each element multiple times.
- 4. Apply performance ratings.
- 5. Add allowances for delays and fatigue.
- **Example Calculation:**
- Task time (observed): 4 minutes.
- Performance rating: 110%.
- Allowance: 10%.
- Standard time = (Observed Time × Rating) + Allowance.

Allowances in Time Study

- **Types of Allowances:**
- - Personal needs.
- - Fatigue.
- - Unavoidable delays.
- **Numerical Example:**
- Observed time: 5 min; Allowance: 15%.
- Standard time = 5 + (5 × 0.15).

$5 \times 1 + 5 \times 0.15 = 5$

$5(0.85) = 5$

$S = \frac{5}{0.85}$

Work Sampling

- **Definition:** Observing random samples to estimate proportions of time spent on activities.
- **Procedure:**
- 1. Define activity categories.
- 2. Take random observations.
- 3. Calculate activity proportions.
- **Numerical Example:**
- Out of 100 observations, 60 were productive. Productivity = 60%.

Practical Applications

- **Case Study:**
- Improving material handling in a manufacturing unit by optimizing transport routes using method study.
- **Before:** 5 steps totaling 10 minutes.
- **After:** 3 steps totaling 6 minutes.
- **Savings:** 40% reduction in time.

A teacher has set a question paper and asked five students to solve it.
Time taken by the students and their corresponding CPIs are given in the table.
Assign a performance rating in the following manner:
1.5 (CPI 9–10), 0.6 (CPI 5–6);
performance rating is a linear function of CPI in the
CPI range 6–9. Providing 10% allowance on the standard time,
find out the standard time to solve the question paper for each student.
What is average standard time and 95% confidence interval?

Student	CPI	Time taken (minutes)
Rahul Kumar	9.27	120
Rahul Singh	8.6	140
Rayansh Jain	5.4	180
Rishi Kawa	6.2	180
Rishi Shaw	9.45	130

Student	CPI	Time (minutes)	P_R	Normal time (minutes)	Standard time (minutes)
Rahul Kumar	9.27	120	1.5	180	180/(1–0.1)=200
Rahul Singh	8.6	140	1.38	193.2	214.67
Rayansh Jain	5.4	180	0.6	108	120
Rishi Kawa	6.2	180	0.66	118.8	132
Rishi Shaw	9.45	130	1.5	195	216.67

Average standard time=176.67 minute

Standard deviation of the standard times=

$$\sqrt{\frac{(176.67-200)^2+(176.67-214.67)^2+(176.67-120)^2+(176.67-132)^2+(176.67-216.67)^2}{4}}$$

$$=\sqrt{\frac{23.33^2+38^2+56.67^2+44.67^2+40^2}{4}}=46.89$$

T-Test Statistic Table – Two-Tailed Significance						
Degrees of Freedom (n-1)	α = 0.20	0.10	0.05	0.02	0.01	0.002
1	3.078	6.314	12.706	31.821	63.657	318.300
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	3.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.305	4.032	5.893

For degree of freedom 4 at 95% confidence, the t-value is 2.776.

Hence, upper limit of the standard

time=176.67+2.776×46.89/2.236=176.67+58.21=234.88

Lower limit of the standard

time=176.67−2.776×46.89/2.236=176.67−58.21=118.46

Confidence interval = (118,235) minutes